SCHOLAR SYSTEM • TECHNICAL SPECIFICATION DOCUMENT

Algorithm Validation & Dataset Requirements Specification

A Comprehensive Blueprint for Empirical Validation, Benchmark Dataset Construction, and Evaluation Metrics

Target System: SCHOLAR Platform | **Matching Algorithms:** TOPSIS + Gale-Shapley | **Focus:** Accuracy, Stability & Integrity

1. Executive Summary & Purpose of Algorithm Validation

In academic research and capstone defense, proposing an algorithm is insufficient without rigorous empirical validation. To prove that the **SCHOLAR** hybrid matching engine functions correctly, the development team must demonstrate that:

- **Hard Filtering Precision:** 100% of ineligible candidates (by residency, education level, or poor academic standing) are barred from receiving awards, with zero false admissions.
- **Ranking Consistency (TOPSIS):** The multi-criteria ranking aligns strongly with human domain experts (e.g., QCYDO scholarship coordinators) using objective multi-attribute scoring.
- **Allocation Stability (Gale-Shapley):** Slot allocations are mathematically stable, respect exact quota capacities, prevent dual-awarding, and resolve ties deterministically without blocking pairs.
- **Authenticity & Explainability:** Document verification (BERT) reliably distinguishes legitimate records, and SHAP explainability provides transparent mathematical justifications.

2. Dataset Specifications: What the System Needs

Validating the recommendation and matching pipeline requires **two operational datasets** (Applicants and Scholarships) and **one ground-truth validation set** (the human expert benchmark).

2.1 Dataset A: The Student / Applicant Dataset (`students.csv / JSON`)

Represents the applicant pool. To evaluate candidates against municipal guidelines, each student record must contain the following standardized features:

Attribute	Data Type	Example Values	Validation Purpose / Algorithm Role
`location` / `is_qc_resident`	String / Boolean	"Quezon City", "Marikina City"	Universal residency filter. Non-residents must be eliminated immediately during pre-screening.
`educationLevel`	Enum String	"College / Undergraduate", "Senior High School", "Postgraduate"	Hard level gating. Prevents college students from matching with Senior High School programs and vice versa.
`yearLevel`	String	"1st Year", "Grade 11", "Graduating"	Gating for cohort-restricted programs (e.g., freshmen-only or graduating grants).
`gpa` / `gwa`	Float (1.00 - 5.00)	1.25, 1.75, 2.50, 3.50	**Critical:** Validates inverted Philippine scale (1.00 is best, 5.00 is failure). Lower numerical value must rank higher.
`percentageGrade`	Float (0 - 100)	88.5, 92.0, 95.0	Validates high school percentage scale (where higher numerical values are better).
`fieldOfStudy`	String	"Computer Science", "Nursing", "Teacher Education"	Evaluates alignment with CHED priority courses and specific scholarship course requirements.
`incomeCategory`	String	"Under ₱25,000", "₱25,000–₱50,000", "Above ₱50,000"	Evaluates poverty threshold compliance for economic and need-based scholarships.
`financialNeed`	Integer (1 to 5)	1 (Lowest Need) to 5 (Urgent Need)	Normalized in TOPSIS to calculate the financial need closeness coefficient.
`isScholarOfAnotherLGU`	Boolean	true / false	Tests the double-dipping restriction (students cannot hold two active government scholarships).
`specialCategories`	Object (Booleans)	isIndigent, isPWD, isSoloParent, isAthlete, isArtist, isStudentLeader	Affirmative action tie-breaker and sector qualifications (10% weight contribution).
`submittedDocuments`	Array of Files	TOR, COE, QCitizen ID, Indigency	Measures document completeness ratio and feeds text to BERT for authenticity confidence scoring.

2.2 Dataset B: The Scholarship Catalog Dataset (`scholarships.csv / JSON`)

Represents the active scholarship programs and partner sponsors against which students compete:

Attribute	Data Type	Example Values	Validation Purpose / Algorithm Role
`name` / `id`	String	"College Academic Scholarship"	Program identifier and receiver entity in Gale-Shapley matching.
`capacity` / `slots`	Integer	3 slots, 5 slots, 50 slots	**Quota constraint:** Tests that Gale-Shapley never allocates more students than available slots.
`requiredEducationLevel`	Array of Strings	["College / Undergraduate"]	Eligibility pre-filter criteria.
`minimumGPA`	Float	1.75 (College) or 89% (SHS)	Cutoff threshold. Any applicant with GWA > 1.75 must be rejected.
`amount`	Currency/Float	₱10,000 / semester	Benefit criterion in TOPSIS; higher monetary values contribute to higher preference.
`requiredDocuments`	Array of Strings	["TOR", "COE", "QCitizen ID"]	Evaluates completeness score: (Submitted ÷ Required) × 100.

2.3 Dataset C: The Ground Truth Benchmark ("Answer Key")

An algorithm cannot be proven correct without an authoritative benchmark. The research team requires:

- **Expert Eligibility Ground Truth:** A binary matrix (Eligible = 1, Ineligible = 0) manually validated by a QCYDO scholarship officer for every student-scholarship pair.
- **Expert Ranking Ground Truth:** A cohort of 10–20 eligible candidates manually ranked from 1st to 20th by scholarship administrators based on official guidelines.
- **Stability Baseline:** Preference tables checked with the algorithm's built-in `isStableManyToOne()` function to confirm zero blocking pairs.

3. Five Essential Test Personas for Edge-Case Validation

To ensure comprehensive coverage, your validation cohort must include these five specific synthetic applicant personas:

Persona 1: The Star Applicant (Positive Control) — Expected: Eligible • Rank #1

Attributes: GWA 1.25, QC Resident, College 2nd Year, Indigent family, 100% complete documents.

Verification: Must clear all pre-screening filters, obtain high TOPSIS closeness score (~93+), and be held in Gale-Shapley priority.

Persona 2: The Inverted GPA Boundary Case

Expected: Ineligible (Failed by 0.01)

Attributes: GWA 1.76, QC Resident, College, complete documents, applying for College Academic (Cutoff: 1.75).

Verification: Tests boundary precision on the Philippine scale. Must be disqualified because 1.76 > 1.75.

Persona 3: The Universal Disqualifier (Early Exit)

Expected: Ineligible across all programs

Attributes: GWA 1.00 (Flawless), High Need, but location is "Marikina City" or has active scholarship in another LGU.

Verification: Proves that exceptional grades cannot override mandatory residency or double-funding restrictions.

Persona 4: The Scale Switcher (SHS vs College)

Expected: Ineligible for SHS Academic

Attributes: Grade 12 student with percentage score of 88.0% applying for SHS Academic (Cutoff: 89.0%).

Verification: Proves the system handles percentage scale where higher is better, without confusing it with the 1.0–5.0 scale.

Persona 5: Over-Quota Capacity & Deterministic Tie

Expected: Exactly 3 Awardees, 2 Rejected

Attributes: 5 qualified applicants with identical match scores competing for a scholarship with capacity = 3.

Verification: Confirms quota is strictly respected (no over-allocation) and ties are broken deterministically by ID.

4. Quantitative Validation Metrics & Formulas

Your thesis manuscript (Chapter 5) requires formal mathematical metrics to prove algorithm accuracy:

4.1 Classification Metrics (Eligibility Pre-Screening)

$$\text{Precision} = TP / (TP + FP) \quad \text{Recall} = TP / (TP + FN) \quad \text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Target Benchmark: Precision = 1.00, Recall = 1.00. Zero false positives ensures that no unqualified student ever receives public funds.

4.2 Rank Correlation Metric (TOPSIS vs Expert Manual Ranking)

To prove that TOPSIS ranks eligible applicants in the same order as a human scholarship board, compute **Spearman's Rank Correlation (r_s)**:

$$r_s = 1 - [(6 \times \sum d_i^2) / (n \times (n^2 - 1))]$$

Where d_i is the difference between the algorithm's rank and the expert's rank for applicant i, and n is the sample cohort size.

Benchmark: An $r_s \geq 0.85$ represents strong correlation. Your benchmark study recorded $\mathbf{r_s = 0.91}$.

4.3 Allocation Stability Metric (Gale-Shapley)

A matching μ is stable if and only if there exists no blocking pair (Student s, Scholarship c) where:

$$\text{Student } s \text{ prefers } c \text{ over } \mu(s) \quad \text{AND} \quad \text{Scholarship } c \text{ prefers } s \text{ over one of its accepted students in } \mu(c)$$

Verification: Enforced via `isStableManyToOne() === true`.

5. How to Run Algorithm Validation in Your Codebase

You can execute the automated test suites already implemented in your project directory:

Command 1: Multi-Criteria Ranking & Gale-Shapley Stability Tests

```
cd Backend ; node --test test/topsis.test.js test/galeShapleyManyToOne.test.js
```

Tests benefit vs. cost criteria, quota capacity adherence, and executes the mathematical stability verifier.

Command 2: Hard Rule Eligibility Tests (Test Cases A to F)

```
cd Backend ; node tests/eligibilityTest.js
```

Evaluates residency gates, GWA cutoffs on the Philippine scale, indigent scoring, and dual-scholarship exclusions.

6. Compliance with Philippine Data Privacy Act (RA 10173)

Notice on Synthetic Benchmark Datasets

Under the Philippine **Data Privacy Act of 2012 (RA 10173)**, publishing or testing with unredacted student transcripts, IDs, and tax documents is prohibited without explicit consent. In academic research, creating and using a **Synthetically Generated Benchmark Dataset** (e.g., 50 anonymized student profiles modeling the statistical distribution of QC applicants) is fully recognized and accepted by thesis defense committees.